



RAJIV GANDHI UNIVERSITY OF KNOWLEDGE TECHNOLOGIES, BASAR
Basar, Nirmal (Dist), Telangana - 504107

(A.Y. 2025-2026)
Branch: Civil Engineering

R22

Subject Name: Building materials and concrete technology

Subject Code: CE2102

Date: 06/08/2025

Time: 45 Minutes

Exam: E2S1MT1

Student ID No: B220367

Max. Marks: 15M

Note:

- Answer all questions from Section -A. Each question carries 1 mark.
- Answer any two questions from Section - B. Each question carries 5 marks.
- If data is found to be missing, assume suitable necessary data.

Section-A

5×1=5M

1. A _____ is a substance which acts as a catalyst and accelerates the process of drying of a paint film. 1M
2. The process of coating the timber with coal tar while it's hot is called _____. 1M
3. The permissible safe load of Ashlar granite in lime mortar is _____. 1M
4. When the bricks are arranged at an angle of 45° from central line in both the directions is called _____. 1M
5. _____ is the development of a series of hair cracks in an irregular pattern over the finished plastering. 1M

Section-B

2 × 5 = 10M

6. a) Explain three types of classification of rocks in detail? 3M
b) Describe any 5 types of Brick Bonds in detail with a neat sketch? 2M
7. a) Describe Bull Trench's Kiln and Hoff mans kiln in detail with a neat sketch? 3M
b) Explain any 5 types of seasoning of Timber in detail? 2M
8. a) Define plastering & pointing ? What are the Defects in plastering ? Explain types of pointing in detail with sketches ? 3M
b) Define (i) paint (ii) varnish (iii) Distemper ? Mention ingredients used in paint in detail? 2M

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(A.Y. 2025-2026)
Branch: Civil Engineering

R22

Subject Name: Building materials and concrete technology

Subject Code: CE2102

Date: 25/09/2025

Time: 45 Minutes

Exam: E2S1MT2 Student ID No: B220367

Max. Marks: 15M

Note:

- Answer all questions from **Section -A**. Each question carries **1 mark**.
- Answer any two questions from **Section - B**. Each question carries **5 marks**.
- If data is found to be missing, assume suitable necessary data.

Section-A

5×1=5M

- What is bouge's compound and its chemical compositions? 1M
- The chemical name for 'ettringite' is _____. 1M
- _____ is one of the most popular equipment for transporting concrete over a long distance. 1M
- The _____ of aggregate is the percentage by weight of particles in it whose least dimension is less than 3/5 th of mean dimension. 1M
- Percentage of bulking formulae = _____ 1M

Section-B

2 × 5 = 10M

- a) Explain the dry process of manufacturing of Ordinary Portland cement in detail with flow chart? 3M
- b) What is heat of hydration? 2M
- a) What is alkali aggregate reaction and what are the factors promoting the alkali aggregate reaction and how to control it in detail? 3M
- b) What is soundness of aggregate? 2M
- a) Explain workability and what are the factors affecting workability? 3M
- b) Define batching and methods of batching in detail? 2M

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(A.Y. 2025-2026)
Branch: Civil Engineering

R22

Subject Name: Building materials and concrete technology
Date: 31/10/2025
Exam: E2S1MT3 Student ID No:

Subject Code: CE2102
Time: 45 Minutes
Max. Marks: 15M

Note:

- Answer all questions from Section -A. Each question carries 1 mark.
- Answer any two questions from Section - B. Each question carries 5 marks.
- If data is found to be missing, assume suitable necessary data.

Section-A

5×1=5M

- _____ is the temperature of the datum line for computing the maturity of concrete. 1M
- What is the Abram's classic law formula? 1M
- The permeability of gel pores in cement paste is estimated to be _____. 1M
- Use of super plasticizers permits reduction of water to _____ percent. 1M
- According to the IS 10262 code book what is the formula for finding the weight of coarse aggregate? 1M

Section-B

2 × 5 = 10M

- a) Calculate the gel/space ratio and the theoretical strength of a sample of concrete made with 500 gm of cement with 0.5 water/cement ratio, on full hydration and at 60% hydration? 3M
- b) What is shrinkage? Explain types of shrinkage in detail? 2M
- a) What are the chemical causes of cracks in concrete after hardening? Explain them in detail? 3M
- b) What is silica fume and how does it affect workability in concrete? 2M
- Design a concrete mix for a construction of an elevated water tank in ACI method. The specified strength of concrete is 30Mpa at 28 days measured on standard cylinders. Standard deviation can be taken 4 Mpa. The specific gravity of FA and CA are 2.65 and 2.7 respectively. The dry rodded bulk density of CA is 1600 kg/m³, and fineness modulus of FA is 2.80. OPC (Type I) is used. A slump of 50mm is necessary. CA is found to be absorptive to the extent of 1% and free surface moisture in sand is found to be 2%. Assume any other data essential. 5M

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(A.Y. 2025-26)

Branch: Civil Engineering

R22

Subject Name:	Building Materials and Concrete Technology	Subject Code:	CE2102
Date:	17-11-2025	Time:	180 Minutes
Exam:	E2SEM-1EST	ID No:	B220367
		Max. Marks:	60

Answer any five of the following questions.

5×12=60Marks

All the answers to the sub questions must be written together. If any data found to missing, assume necessary suitable data.		
1. a)	What is brick masonry? Describe various types of bonds in brick masonry.	6M
b)	What is meant by seasoning of timber? Discuss in detail about the various defects in timber with neat sketches.	6M
2. a)	Distinguish between plastering and pointing.	6M
b)	Explain in detail about the faults in painting work with an example.	6M
3. a)	Explain the roles of the various components of cement and the influence of their hydration products on the properties of hardened cement.	6M
b)	What is an alkali aggregate reaction? Mention its effect on concrete structures.	6M
4. a)	What are the various factors affecting workability of fresh concrete mix? Explain in detail.	6M
b)	Discuss briefly about the following – Segregation and bleeding of concrete.	6M
5. a)	Explain about the following terms: i) Gel space ratio ii) Maturity concept of concrete	6M
b)	Discuss in detail about Creep and Shrinkage effects in concrete.	6M
6. a)	Differentiate between chemical and mineral admixture.	6M
b)	State the requirements of Physical and Mechanical properties of coarse aggregates in the process of concrete preparation?	6M
7. a)	Design a M40 concrete using IS 10262 for the following data. Maximum size of aggregate = 20 mm (angular, IS 383), Compacting factor 0.93, Type of exposure – Moderate, Specific gravity of Cement, FA, CA = 3.15, 2.6, 2.7, Zone of sand = Zone III, Water absorption of CA, FA = 0.6%, 1.2%, Free moisture on CA, FA = 0.1%, 2.0%.	12M

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